

A Discontinuous Conduction Mode Single-Stage Step-Up Rectifier for Low-Voltage Energy Harvesting Applications

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Abstract—In this paper, a novel ac/dc converter is proposed for low-voltage and low-power ac rectification applications. The proposed converter manages the energy harvested from microscale electromagnetic generators (MEGs). It integrates the conventional boost and buck–boost topologies with a shared inductor, a bidirectional switch, and two split filtering capacitors. The boost and the buck–boost topologies function in the positive and negative half input cycles, respectively. The inductor is energized by being shorted with the input source through the MOSFET channel without using the diodes. This enables active rectification of low-amplitude (below 0.7 V) ac voltages. Theoretical analysis, design considerations, and control method are detailed. A 40-mW circuit prototype, which converts a 0.4-V peak, 100-Hz ac voltage source to 3.3-V dc, is designed and tested to verify the proof of concept.

Index Terms—AC/DC conversion, boost, buck–boost, energy harvesting, low power, low voltage.

I. INTRODUCTION

IN RECENT years, low-voltage and low-power electronic devices for a variety of applications such as wireless sensor node and wearable devices have been increasingly reported in the literature [1]–[7]. In general, kinetic energy is generated from ambient vibrations and random displacements of mechanical and biological systems [8]. Electromagnetic, piezoelectric, and electrostatic transducers are commonly employed to convert the kinetic energy into electricity [9]. Microscale electromagnetic generators (MEGs) are suitable candidates to supply power to low-power self-sustainable electronic devices due to their better power densities [10].

An MEG consists of a coil, a permanent magnet (PM), and a spring [11], [12]. Ambient vibration introduces the relative displacement between the PM and the coil. The coil cuts through

the magnetic flux alternatively and induces an ac voltage output. Szarka *et al.* [13] show a vibration-based MEG with more than 1-mW power capability at its resonant frequency of 59.4 Hz. In [14], a 35-mW MEG at its resonant frequency of 42 Hz is reported. Dayal *et al.* [15] reported an MEG used as ac energy harvesting source with a resonant frequency of 108 Hz and 0.45-V output. Due to the size limitation, the output of an MEG is typically a low-amplitude and low-frequency ac voltage [16]. However, most of the electronic devices require a 2–3.3 V dc to operate. Therefore, it is critical to design a power electronic interface, which can efficiently convert this low-amplitude ac voltage to a boosted and regulated dc voltage.

Traditional ac/dc topologies generally consist of two stages [14], [17]: a front stage diode bridge rectifier and a cascaded dc/dc converter. However, the main disadvantages of the two-stage topologies lie in: 1) unavoidable diode voltage drop, which makes it infeasible in low-voltage input applications; and 2) significant power losses in the front-end diode bridges. Multiple direct ac/dc topologies have been reported in recent years. A two-switch H-bridge topology, which integrates two boost converters in low-voltage energy harvesting applications, is proposed in [18]. However, when the switch is OFF, the inductor current continues to flow via two diodes. This deteriorates both the boost ratio and conversion efficiency. A four-switch synchronous boost rectifier is applied to MEG energy harvesting applications in [19]. However, it has higher switch count and requires a complicate and costly polarity detector. Generally, this polarity detector is lossy considering the low power level of the energy harvesting system. However, utilizing ultralow-power integrated circuits would significantly reduce the loss of the polarity detector [20]. Two more direct ac/dc topologies are proposed in [21] and [22]. The circuit in [21] consists of two boost converters; while the circuit in [22] consists of a boost converter and a buck–boost converter. However, both converters require two magnetic inductors and polarity detector.

A much simpler circuit, which integrates boost and buck–boost topologies while utilizing single inductor, is proposed in [23]. However, it also requires a complicated and costly polarity detector. This increases control circuit complexity and power consumption. To address those problems, a direct ac/dc power conditioning circuit, which consists of a boost converter and a buck–boost converter, is proposed in this paper, as shown in Fig. 1. In this topology, there is a bidirectional switch, which is composed of a p -MOSFET and an n -MOSFET. Two MOSFETs are

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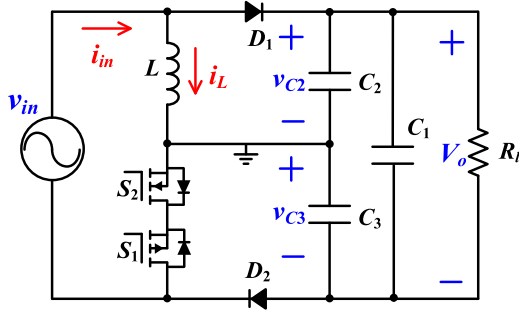


Fig. 1. Schematic of the proposed single-stage rectifier.

turned ON and OFF simultaneously. At the same time, two split capacitors and two diodes are used to realize the operating mode transitions. This topology demonstrates the following benefits:

- 1) reduced circuit size;
- 2) simplified control circuit;
- 3) removal of the complicated and costly polarity detector.

A direct ac/dc power conditioning circuit consisting of two boost converters is reported in [24]. This circuit also utilizes split capacitors and a bidirectional switch.

The operation modes of converter are described in Section II. Detailed theoretical analyses are provided in Section III. Design considerations and procedures are discussed in Section IV. In Section V, the design of the controller and MOSFET driver is presented. The simulation results are presented in Section VI. Experiment results are reported in Section VII. Section VIII concludes the paper.

II. OPERATION PRINCIPLE

It should be noted that the realistic energy harvester might demonstrate considerable source impedance. However, a more ideal source with neglectable source impedance would facilitate the fundamental topology analysis. This paper is mainly focused on the topology analysis. Therefore, an ideal sinusoidal signal source is adopted in the converter analysis, while a realistic signal source is simulated to verify its capability in the real energy harvesting applications. This study considers only the MEG with low output voltage. A 0.4 V and 100-Hz sinusoidal ac voltage is used to emulate the output of the MEG.

As shown Fig. 1, in the proposed topology, only one inductor is utilized. In order to rectify the ac input to a dc voltage, the switch should be able to conduct and block currents in both directions during the ON and OFF states, respectively. Since a MOSFET is a single quadrant switch, i.e., in ON state, MOSFET channel can conduct bidirectional current; while in OFF state, it can only block unidirectional drain to source voltage. This is due to the existence of its body diode. Therefore, a *p*-MOSFET and an *n*-MOSFET are placed in series with their body diodes back to back, to function as a four-quadrant switch. It should be noted that turning ON and OFF of those two MOSFETs is always synchronized.

In the positive half-cycle of the input voltage v_{in} , it operates as a boost converter (including L , D_2 , and C_3). In the negative half-cycle of v_{in} , it operates as a buck-boost converter (including L ,

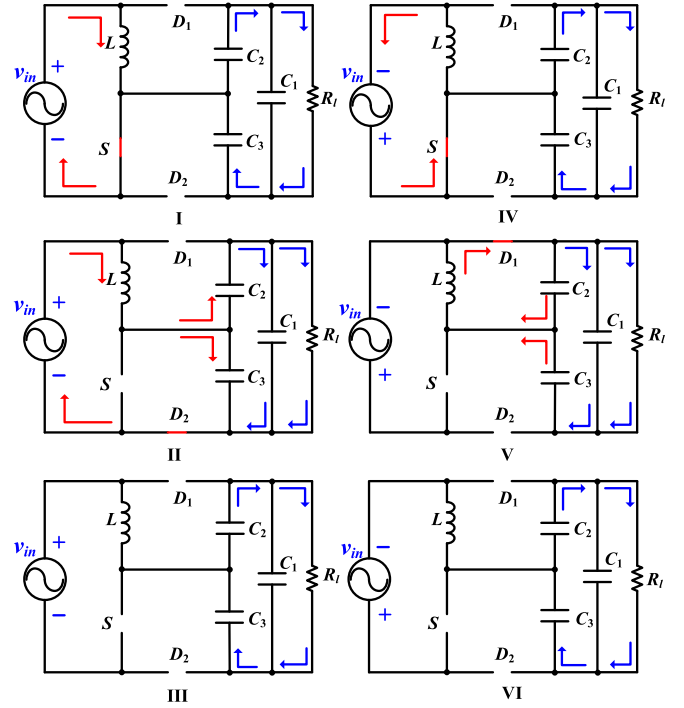


Fig. 2. Operating modes of the proposed converter.

D_1 , and C_2). The discontinuous conduction mode (DCM) operation has the benefits of reduced switching losses and mitigated diode reverse recovery. Therefore, this converter is designed to operate in DCM. In the DCM operation, the converter has three main operating modes in each half-cycle, as shown in Fig. 2. Modes I–III correspond to the positive half-cycle, while Modes IV–VI correspond to the negative half-cycle.

Mode I: When the switch is turned ON, the inductor is energized by the input source. The inductor current increases linearly from zero. The switch is turned ON with zero current. Therefore, switching losses are reduced. Meanwhile, both diodes are reverse biased. The output capacitors supply power to the load.

Mode II: When the switch is turned OFF, i_L freewheels via D_2 . C_3 is charged while C_2 is discharged. The voltage across the inductor equals to the difference between input voltage and lower capacitor voltage (v_{C3}), which is a negative value. Therefore, i_L decreases linearly until it crosses zero.

Mode III: When the i_L becomes zero, D_2 turns OFF naturally with a small di/dt . This significantly mitigates the diode reverse recovery loss. The capacitors discharge themselves through the load. Given the input voltage is still positive, the converter returns to Mode I as soon as the switch turns ON.

Mode IV: Modes IV–VI happen when v_{in} is negative. Mode IV is similar to Mode I as soon as switch turns ON. i_L linearly deviates from zero; the inductor is energized by the input source. The switch is also turned ON at zero current. On the other hand, both diodes are reverse biased. The output capacitors supply power to the load.

Mode V: When the switch is turned OFF, i_L continues to free-wheel via D_1 . C_2 is charged while C_3 is discharged. The voltage across the inductor equals to v_{C3} . Therefore, i_L increases linearly until it crosses zero.

Mode VI: When the inductor current increases to zero, D_1 turns OFF automatically. This avoids the reverse recovery losses of the diode. In this condition, the capacitors supply power to load again. Given the input voltage is still negative, the converter returns to Mode IV.

According to the operating modes analyses, in the DCM operation, both the MOSFETs turning ON and the diodes turning OFF occur at zero current. Therefore, the switching losses are reduced.

III. THEORETICAL ANALYSIS

To maximize the extracted power from MEG, impedance matching between the source impedance and the input impedance of the converter is necessary. It should be noted that the input impedance of the proposed converter is adjustable by tuning the duty cycle and switching frequency. However, this paper is mainly focused on fundamental topology analysis and verification. Therefore, the input of the proposed converter is considered as a sinusoidal ac voltage to simplify the circuit analysis. In the proposed converter, two split capacitors, C_2 and C_3 , are used. Capacitor C_3 is charged and C_2 is discharged in the boost operation; while capacitor C_2 is charged and C_3 is discharged in the buck–boost operation. Assuming that the capacitor is large enough to keep the voltage constant, the voltages of C_2 and C_3 are constant and equal to the half of output voltage. The switching frequency (f_s) is much larger than the frequency of the input source. Therefore, the input voltage could be considered constant during each switching period. Thus, v_{in} can be expressed as follows:

$$v_{in}(t) = V_P \sin(2\pi t/T_{in}) \quad (1)$$

where T_{in} is the period of an input ac source. V_P is the amplitude of v_{in} . The input current is equal to the inductor current in the boost operation. However, in the buck–boost operation, the input current becomes zero when the switch is turned OFF, as shown in Fig. 3. T_S is the switching period, and d_1 is the duty cycle. In the boost mode, the peak value of the input current can be expressed as follows:

$$i_{PK}(t) = d_1 T_S v_{in} / L. \quad (2)$$

According to the inductor volt–second balance

$$v_{in} d_1 T_S = (V_o/2 - v_{in}) d_2 T_S. \quad (3)$$

The average input power in each switching period is derived as follows:

$$p_{in} = v_{in} i_{PK} (d_1 + d_2) / 2. \quad (4)$$

The input energy of the positive half-cycle is derived as follows:

$$E_{in} = \int_0^{T_{in}/2} v_{in} i_{PK} T_S (d_1 + d_2) / 2 dt. \quad (5)$$

According to (1)–(5)

$$E_{in} = \int_0^{T_{in}/2} \frac{v_{in}^2 T_S d_1^2}{2L} \left(1 + \frac{v_{in}}{V_o/2 - v_{in}}\right) dt \quad (6)$$

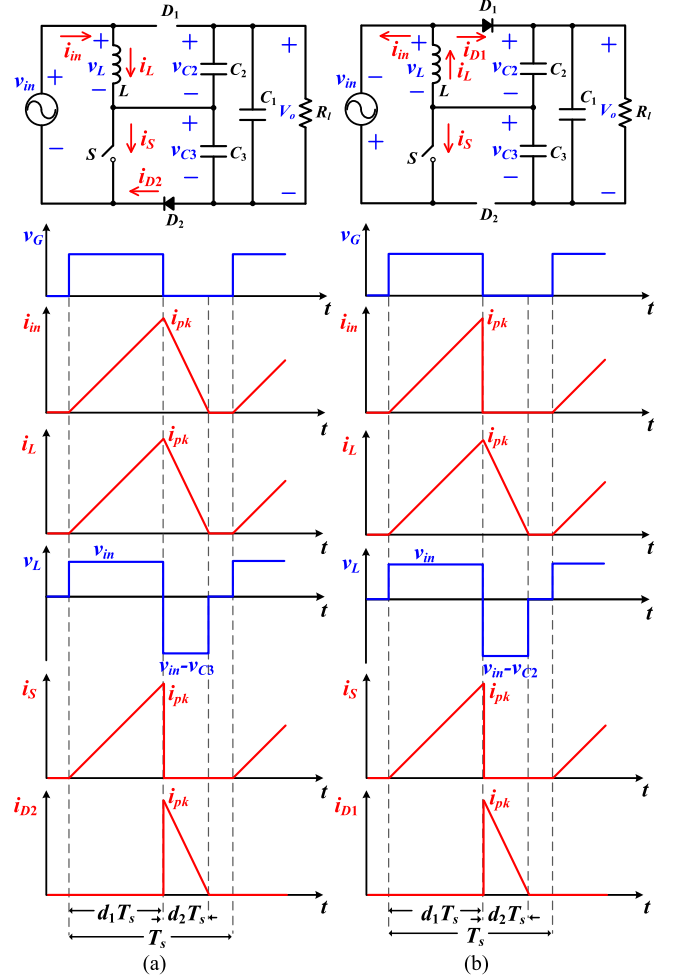


Fig. 3. Waveforms of the proposed boost/buck–boost rectifier.

where $V_o/2$ is much larger than the average value of v_{in} from $0 - T_{in}/2$, so the last item could be ignored. Hence, E_{in} is expressed as follows:

$$E_{in} \approx V_P^2 T_S d_1^2 T_{in} / 8L. \quad (7)$$

According to (7), without considering internal resistance of the ac voltage source, the input energy is inversely proportional to inductance and proportional to square of the duty cycle and the switching period. Assuming all components are ideal, the output energy should be equal to the input energy in the same time scale. The output energy of the positive half-cycle can be derived as follows:

$$E_{out} = \frac{V_o^2 T_{in}}{2R_l} = \frac{V_P^2 T_S d_1^2 T_{in}}{8L} \quad (8)$$

where R_l is the load resistance. Therefore, the output energy can be controlled by the duty cycle and the switching frequency. It should be noted that (8) also applies to the buck–boost mode. This is due to the similar approximation in (7), which can be extended to the buck–boost mode.

IV. DESIGN PROCESS

In order to optimize the performance of the converter, the voltage gain and efficiency should be derived. According to (8),

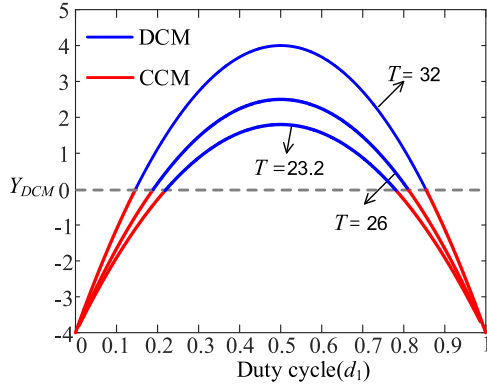


Fig. 4. DCM conditions of the proposed converter.

the voltage gain can be derived as follows:

$$\frac{V_o}{V_P} = \frac{d_1}{2} \sqrt{\frac{T_S R_L}{L}}. \quad (9)$$

According to (9), the output voltage can be controlled by the duty cycle and the switching period. Besides, the converter is supposed to operate in DCM to reduce the switching losses. In the boost mode, the DCM criteria can be expressed as follows:

$$i_{PK} - (1 - d_1) (V_o/2 - v_{in}) T_S / L < 0. \quad (10)$$

To make sure that the converter can operate in DCM under all input voltages, it should be ensured that

$$V_P < (1 - d_1) V_o/2. \quad (11)$$

Based on (9) and (11), the DCM condition can be expressed as follows:

$$d_1 (1 - d_1) \sqrt{T_S R_L / L} - 4 > 0. \quad (12)$$

The same analysis can be used in the buck–boost mode, and the DCM condition can be expressed as follows:

$$(1 - d_1) \sqrt{T_S R_L / L} - 4 > 0. \quad (13)$$

Assume that

$$Y_{DCM} = d_1 (1 - d_1) \sqrt{T_S R_L / L} - 4 \quad (14)$$

$$T = \sqrt{T_S R_L / L}. \quad (15)$$

According to (12) and (13), when the duty cycle is close to zero, the boost converter operates in the continuous conducting mode (CCM) and the buck–boost converter operates in DCM. When the duty cycle is close to one, both the buck–boost converter and the boost converter operate in CCM. Thus, if the Y_{DCM} is positive, the proposed converter must operate in DCM. As shown in Fig. 4, the operating state of converter is determined by the duty cycle, inductance, and switching frequency. Larger T corresponds to wider duty cycle range. T can be kept constant by simultaneously reducing T_S and L . In the DCM operation, the MOSFET is turned ON and the diode is turned OFF with zero current. Thus, the switching loss is trivial in comparison with the conduction loss. This means a relatively high switching frequency may be selected. However, high switching frequency

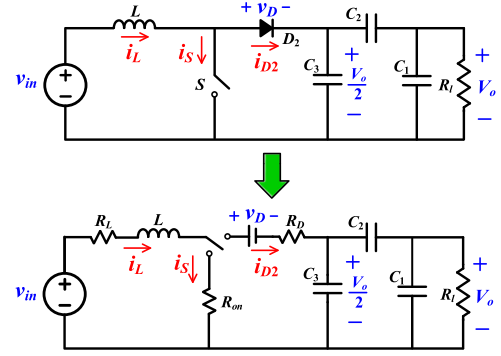


Fig. 5. Boost mode equivalent circuit.

would boost the inductor iron loss, driver loss, as well as the parasitic capacitor loss. Besides, high switching frequency might lead to the CCM operation. Therefore, based on this tradeoff, 50 kHz is selected to verify the circuit operation under DCM with acceptable conversion efficiency.

The efficiency analysis of the converter operating as a boost converter for the positive input voltage is based on the following assumptions:

- 1) In forward bias condition, the diode is modeled as a constant voltage source v_D in series with a resistor R_D . Moreover, the switching losses in the diode are neglected.
- 2) Inductor current varies linearly. The equivalent series resistance R_L of the inductor is independent of its operating temperature.
- 3) Power losses corresponding to the capacitors' equivalent series resistances are neglected.

When the proposed converter operates in the boost mode, the equivalent circuit is shown in Fig. 5. The current waveforms i_L , i_D , and i_S are shown in Fig. 3(a). The rms value of the inductor current over the whole switching cycle can be expressed as follows:

$$I_{L-RMS}^2 = \frac{1}{T_S} \int_0^{T_S} i_L^2(t) dt = \frac{i_{PK}^2 (d_1 + d_2)}{3}. \quad (16)$$

The switch and the diode currents, respectively, can be calculated in the same manner

$$I_{S-RMS}^2 = \frac{1}{T_S} \int_0^{T_S} i_S^2(t) dt = \frac{i_{PK}^2 d_1}{3} \quad (17)$$

$$I_{D-RMS}^2 = \frac{1}{T_S} \int_0^{T_S} i_D^2(t) dt = \frac{i_{PK}^2 d_2}{3}. \quad (18)$$

The inductor average current can be derived as follows:

$$I_L = \frac{1}{T_S} \int_0^{T_S} i_L(t) dt = \frac{i_{PK} (d_1 + d_2)}{2}. \quad (19)$$

The diode average current can be expressed as follows:

$$I_D = \frac{1}{T_S} \int_0^{T_S} i_D(t) dt = \frac{i_{PK} d_2}{2}. \quad (20)$$

For the low frequency, the winding loss is the main loss of inductor. So the inductor loss can be expressed as follows:

$$P_{RL} = R_L I_{L-RMS}^2 = \frac{(d_1 + d_2) R_L i_{PK}^2}{3}. \quad (21)$$

The MOSFET conduction loss is expressed as follows:

$$P_{RS} = R_{on} I_{S-RMS}^2 = \frac{d_1 R_{on} i_{PK}^2}{3}. \quad (22)$$

The diode power loss is

$$P_D = v_D I_D + R_D I_{D-RMS}^2 = \frac{d_2 v_D i_{PK}}{2} + \frac{d_2 R_D i_{PK}^2}{3}. \quad (23)$$

During the turning OFF of the bidirectional switch, the switch's reverse biased voltage changes slowly from zero to a dc value ($V_o/2$) because of its parasitic output capacitor (C_{OSS}). Therefore, the turning OFF loss of the bidirectional switch is also small and can be ignored. The main switching loss is the parasitic capacitors loss. At the beginning of Mode II, the voltage across the bidirectional switch is precharged to $V_o/2$. However, at the beginning of Mode IV, this voltage is discharged to 0 V. Thus, the switching loss of two MOSFETs can be expressed as follows:

$$P_{SW} = \frac{C_{OSS} V_o^2}{8T_s} \quad (24)$$

where C_{OSS} is the switch output capacitance.

The input current is equal to the inductor current. So the input power can be expressed as follows:

$$P_{in} = v_{in} I_{in} = \frac{1}{2} (d_1 + d_2) v_{in} i_{PK}. \quad (25)$$

The efficiency of converter can be expressed as follows:

$$\eta = 1 - \frac{P_{RL} + P_{RS} + P_D + P_{SW}}{P_{in}}. \quad (26)$$

Based on the aforesaid equations, we can get

$$\eta \approx 1 - \left[\frac{2d_1 T_s (R_L + R_{on})}{3L} - \frac{4R_{on}}{3} \sqrt{\frac{L}{R_L T_s}} + \frac{4v_D}{V_P d_1} \sqrt{\frac{L}{R_L T_s}} + \frac{C_{OSS}}{T_s} \left(\frac{R_L}{4} - \frac{1}{2d_1} \sqrt{\frac{R_L L}{T_s}} \right) \right]. \quad (27)$$

Equation (27) does not contain V_o . This is because V_o is represented by the other variables according to (9). According to (27), the greater R_L , R_{ON} , and V_D are, the smaller the efficiency is. The larger V_P leads to a higher efficiency. With R_L , R_{on} , V_D , and C_{OSS} equal to 0.09 Ω , 0.1 Ω , 0.25 V, and 60 pF, respectively, the relationship between efficiency and duty cycle is illustrated in Fig. 6. According to (9) and assuming that f_s , R_L , and d_1 are all constant, larger L corresponds to smaller output power. This means that larger L corresponds to a smaller power level. According to Fig. 6, the larger L corresponds to higher efficiency when the duty cycle is larger than 0.65. However, based on Fig. 4, the decrease of L might induce the DCM operation. Thus, the tradeoff among output power, DCM, and T should be considered. In this study, the output power is 40 mW. T is equal to 23.2 ($R_L = 270 \Omega$, $f_s = 50$ kHz, $L = 10 \mu\text{H}$). The power losses distribution is plotted in Fig. 7. It should be noted

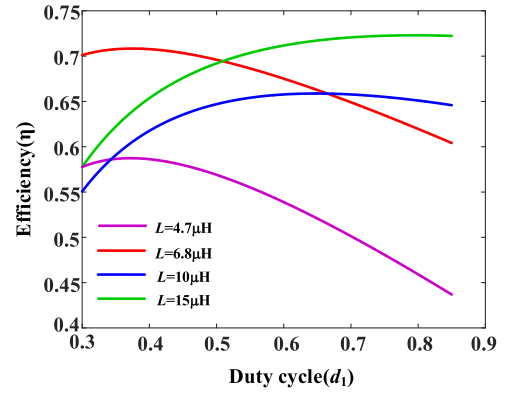


Fig. 6. Theoretically predicted efficiency versus duty cycle with different L ($R_L = 270 \Omega$, $f_s = 50$ kHz).

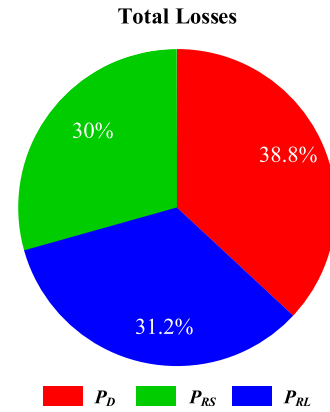


Fig. 7. Power losses breakdown ($T = 23.2$, $d_1 = 0.75$).

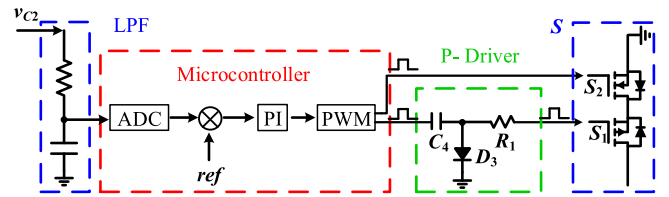


Fig. 8. Controller and gate driver of the proposed converter.

that the switching losses P_{SW} much smaller than the other losses can be neglected.

V. CONTROL STRATEGY

The proposed converter should stably rectify a 0.4-V amplitude ac voltage and boost it to a 3.3-V dc voltage. Thus, a voltage feedback control loop is necessary to regulate the output voltage. The controller block diagram is plotted in Fig. 8. The output voltage equals the sum of v_{C2} and v_{C3} . It is feasible to estimate C_1 's voltage by sampling C_2 's voltage. Then, the controller is able to tune the duty cycle of the four-quadrant switch to maintain a constant output voltage on C_1 .

In the hardware implementation, a low-pass filter (LPF) is used to calculate C_2 's average voltage. Then, this analog signal

is fed to the analog-to-digital converter (ADC) of the digital controller. In this paper, a simple classical PI algorithm is adopted to achieve constant voltage output. P and I are selected as 3 and 0.1, respectively. The closed-loop experiment is conducted to verify that the converter is able to demonstrate good stability performance. The microcontroller with on-chip pulse width modulation (PWM) module, integrated ADC, and high clock frequency is adopted in this paper. In this paper, a p -MOSFET and an n -MOSFET are put in series to form the four-quadrant switch. The switching of S_1 and S_2 is always synchronized without polarity detection of the input voltage.

The driving of S_2 (n -MOSFET) is straightforward. One can directly apply the PWM output of the controller to the gate and source of the MOSFET to turn ON and OFF S_2 . However, driving of S_1 (p -MOSFET) is more complicated since its turning ON requires a negative voltage. Thus, a driver circuit is required. A simple driver circuit, which composes of a capacitor, a resistor, and a diode, as shown in Fig. 8, is used. R_1 is used to limit the gate current and selected to be $3.3\ \Omega$. C_4 should be large enough to keep its voltage constant. In this paper, $10\ \mu\text{F}$ is selected for C_4 . When the gate signal is high, C_4 is charged through the diode and R_1 . V_{GS1} is positive and equal to the forward bias voltage of diode. Therefore, the p -MOSFET is turned OFF. When the gate signal is low, C_4 is slightly discharged through R_1 . V_{GS1} is negative and equal to $-v_{C4}$. Then, the p -MOSFET turns ON.

VI. SIMULATION RESULTS

Fig. 9(a) shows the circuit simulation waveforms using an ideal signal source with zero source impedance. While Fig. 9(b) shows the simulation waveforms using a more realistic irregular signal source with an unneglectable source impedance. As can be seen, the proposed topology is able to rectify the ac output of a realistic energy harvester to a stable dc voltage.

VII. EXPERIMENT RESULTS

In order to minimize the size of proposed circuit, surface-mounted components are adopted. A highly compact power electronic interface prototype is fabricated as shown in Fig. 10. The prototype has a PCB size of $3 \times 2.5\ \text{cm}^2$. Schottky diodes with low forward voltage drop are used. To minimize the conduction losses, low voltage rating MOSFETs with low ON resistances are adopted. The n -MOSFET is IRLML2502, and the p -MOSFET is IRLML6401. Tantalum capacitors are selected as the filtering capacitors since they have small leakage current and compact size. In this study, the MEG is emulated by a signal generator cascaded with a power amplifier. The ac power source is designed to have 0.4-V amplitude and 100-Hz frequency. The output voltage is 3.3-V. The designed prototype achieves 60% conversion efficiency at 40 mW. The duty cycles of the converter are about 0.75 at 50-kHz switching frequency. The ratings of selected components are shown in Table I.

The experiment results are shown in Figs. 11–19. As seen from Fig. 11, the input voltage is a 0.4 V and 100-Hz sinusoidal ac voltage. The inductor current envelop is synchronized with the input voltage at the steady state. The voltages of the split capacitors are shown in Fig. 12. The sum of two capacitors'

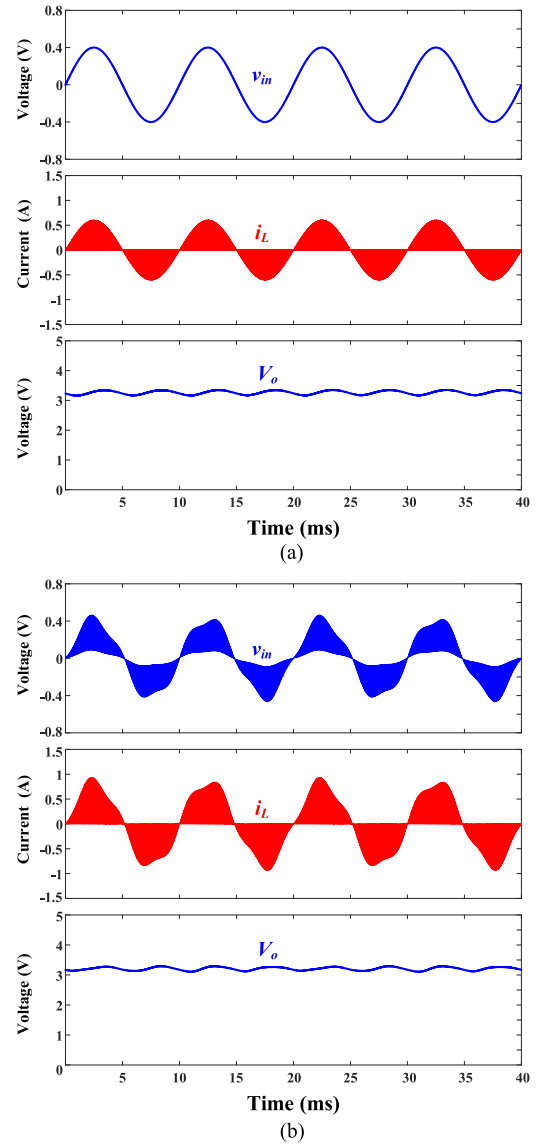


Fig. 9. Converter simulation results with (a) ideal sinusoidal input and zero source impedance; and (b) irregular input and source impedance.

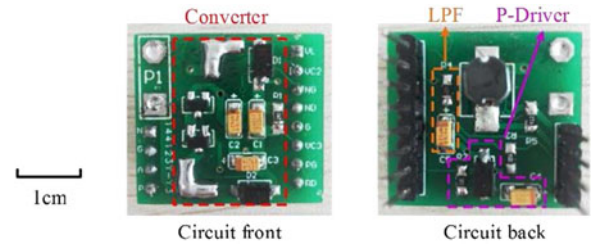


Fig. 10. $3 \times 2.5\ \text{cm}^2$ prototype of the proposed converter.

voltages is equal to the output voltage. Fig. 13 shows the 3.3-V output voltage of proposed converter with a small ripple.

The inductor current for the boost mode operation is shown in Fig. 14. The DCM operation can be observed. Fig. 15 shows the input current and gate voltage in the boost operation. The input current is equal to the inductor current. However, the input

TABLE I
PARAMETERS OF COMPONENTS

Component	Symbol	Parameters
Inductor	L	$10\ \mu\text{H}$
Split capacitor	C_2, C_3	$47\ \mu\text{F}$
Output capacitor	C_1	$100\ \mu\text{F}$
N-channel MOSFET	S_2	20 V, 4.2 A
Load	R_L	$270\ \Omega$
P-channel MOSFET	S_1	20 V, 4.2 A
Schottky Diodes	D_1, D_2	40 V, 3 A

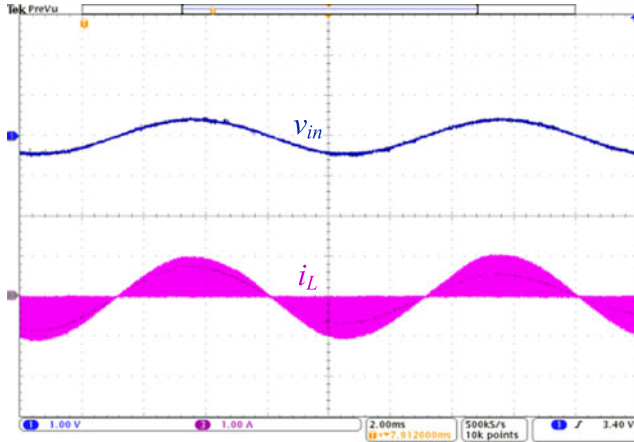


Fig. 11. From top to bottom: input voltage (1 V/div), inductor current (1 A/div); time 2 ms/div.

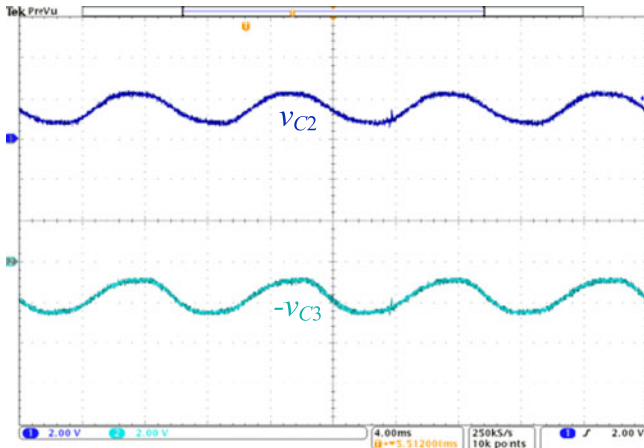


Fig. 12. Voltages on the split capacitors (2 V/div); time 4 ms/div.

current is equal to zero in the buck–boost operation in Mode V, as shown in Fig. 16. These results agree with previous theoretical analysis. As shown in Fig. 17, the reverse recovery process of the diode is avoided.

According to the theoretical analysis, this converter operates in CCM with the duty cycle equal to 0.85. This is verified experimentally: Fig. 18 clearly shows that the circuit enters into CCM with a duty cycle equal to 0.85. Closed loop voltage control is implemented, as shown in Fig. 19. When the load resistant changes from 500 to 270 Ω , the system adjusts quickly to

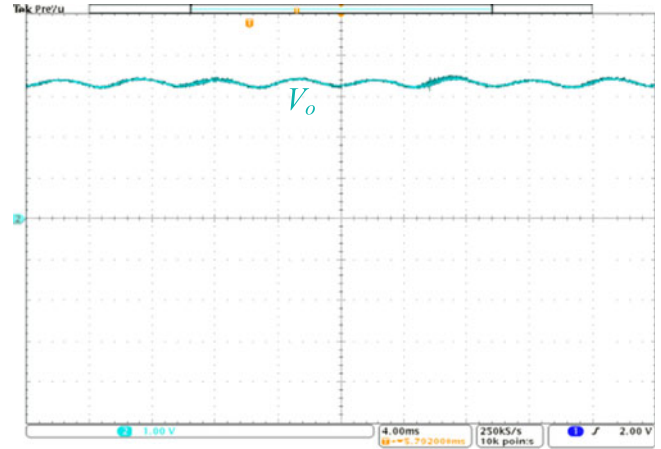


Fig. 13. Output voltage (1 V/div); time 4 ms/div.

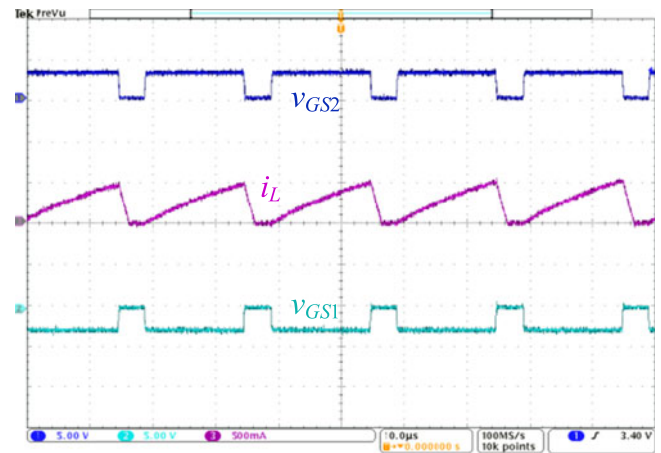


Fig. 14. Circuit waveforms in the boost mode operation, from top to bottom: gate source voltage of S_2 (5 V/div), inductor current (0.5 A/div), gate source voltage of S_1 (5 V/div); time 10 μs /div.

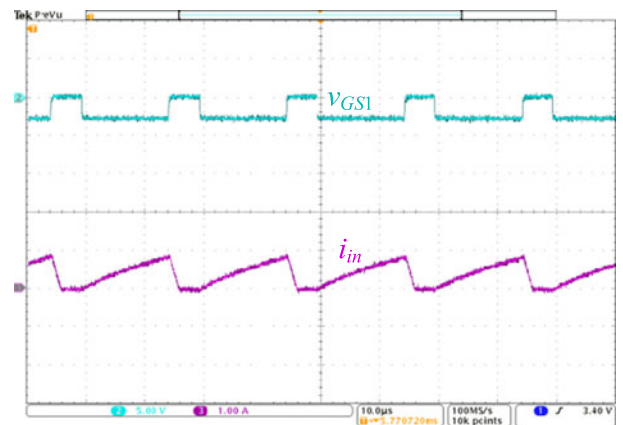


Fig. 15. From top to bottom: the gate source voltage of S_1 (5 V/div), input current (1 A/div); time 10 μs /div.

maintain a constant output voltage. The design converter is able to rectify a 0.3–1.8 V, 20–5 kHz ac voltage to a stable 3.3-V dc voltage. The rated output power is 67 mW. The losses of individual component are listed in Table II. The experimental results are similar to the previous theoretical analysis. A comparison

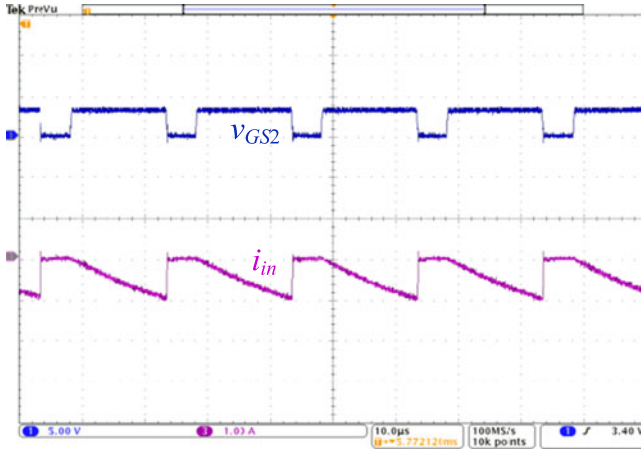


Fig. 16. From top to bottom: the gate source voltage of S_2 (5 V/div), input current (1 A/div); time 10 μ s/div.

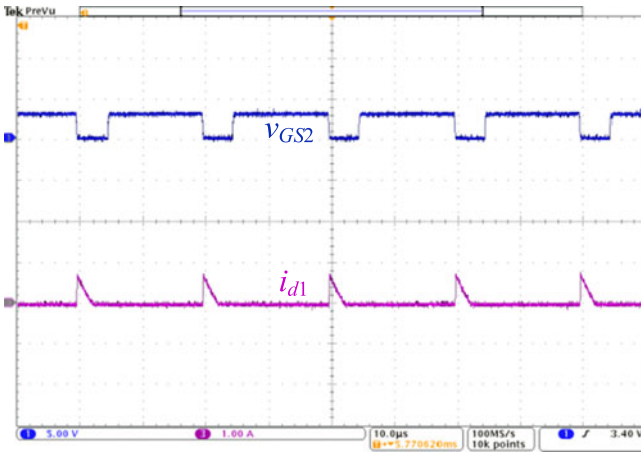


Fig. 17. From top to bottom: the gate source voltage of S_2 (5 V/div), diode current (1 A/div); time 10 μ s/div.

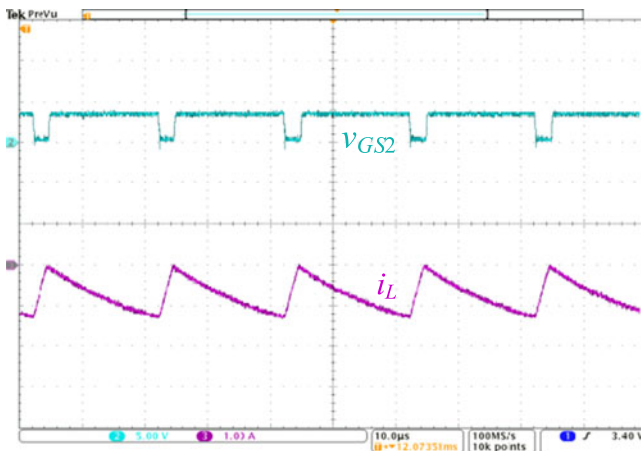


Fig. 18. From top to bottom: the gate source voltage of S_2 (5 V/div), inductor current (1 A/div); time 10 μ s/div. Proposed converter operates in CCM (duty cycle 0.85).

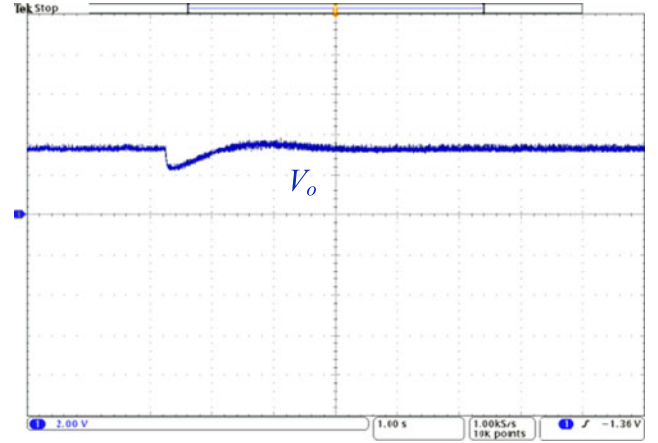


Fig. 19. Output voltage of proposed converter with closed loop control (2 V/div); time 1 s/div.

TABLE II
LOSSES MEASUREMENT

Category	Component	Loss
Power Circuit	Inductor	4.9 mW
	Diodes	6.5 mW
	MOSFETS	4.3 mW
	Capacitors	0.3 mW
Control loop	Driver	5 mW
	LPF	0.2 mW
Total Loss		21.2 mW

TABLE III
PERFORMANCE SUMMARY AND COMPARISON

Ref	f (Hz)	v_{in} (V)	P_{out} (mW)	η (%)	Inductor count	Polarity detector
[21]	322	0.3	50	50	2	yes
[22]	100	0.4	54.5	61	2	yes
[23]	100	0.4	54.5	71	1	yes
[24]	108	0.2–0.8	10	55	1	no
Proposed converter	100	0.4	40	60	1	no

of the proposed converter with other low-power converters is depicted in Table III.

VIII. CONCLUSION

A novel ac/dc topology is proposed in this paper. The proposed single stage converter is able to rectify and step up low amplitude ac voltages. Therefore, it is suitable to be used in low-power MEG energy harvesting applications. This topology integrates the conventional buck–boost converter and boost converter into a single stage with shared energy storage inductor and a four-quadrant switch. The proposed converter successfully avoids the introduction of a complicated and costly polarity detector circuit. The DCM operation is adopted to decrease the switching losses and improve the efficiency. The operation and performance of the proposed converter is validated by

simulation and experiment results. The designed prototype converts a 0.4-V 100-Hz ac input to a stable 3.3-V dc voltage, with a 60% conversion efficiency at 40-mW output power.

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